

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Conference
Manufacturer: Cisco
Firmware Version: CE8.0.0
Model(s): MX800 CE8.0.X, MX700 CE8.0.X

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.05.0000-b007	2.01.0002-b005	1.3.0

Version History

Module Version	Date	Notes
1_1_0_1	6/16/2017	Fix notes in comm sheet. Fixed commands: Standby, PhonebookUpdate, CallID.
1_1_0_0	3/17/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- Use highest baud rate for better performance when using serial control.
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5
- CallHistoryOccurrenceType variable must be set accordingly. Default value is 'Time'. Call History Occurrence Type can be set to 'Time' or 'Frequency'. Example: InterfaceName.CallHistoryOccurrenceType = 'Time'
- NumberofCallHistory variable must be set accordingly. Default value is '5'. Number of Call History ranges from '1' to '20'. Example: InterfaceName.NumberofCallHistory = '1'
- NumberofPhonebookFolderSearch variable must be set accordingly. Default value is '5'. Number of Phonebook Folder Search ranges from '1' to '10'. Example:
InterfaceName.NumberofPhonebookFolderSearch = '1'
- NumberofPhonebookSearch variable must be set accordingly. Default value is '5'. Number of Phonebook Search ranges from '1' to '10'. Example: InterfaceName.NumberofPhonebookSearch = '1'

Phonebook commands:

- PhonebookSearch
 - This command takes in a search phrase and it is used to search the contact names that would match this. Setting or appending characters to this command will trigger a search in the contact list respectively to the selected phonebook type.
- PhonebookSearchResult
 - This is where the search results will be saved to when phonebook is searched. The number of results are set by a driver parameter Number of Phonebook Search, this is initialized to 5 results by default. Simply call SetText with the returned result to display the search result on a button or label object.
- PhonebookSearchSet
 - When the number selected in this command is triggered, it will dial the number with the xCommand Dial Number command.
- PhonebookNavigation
 - This command is used for navigating through the search results. The navigation commands scroll through the results when there are more results than NumberofPhonebookSearch variable.
- PhonebookUpdate
 - This is used to trigger the entire Phonebook search.
- PhonebookFolderIDUpdate
 - This is used to trigger the Folder search within the Phonebook.
- PhonebookFolderIDSearchSet
 - When the Folder Name selected in this command is triggered, it will store the Folder Name in a class variable, FolderIDNumber, and that variable can be accessed and used in the 'FolderID' qualifier in the PhonebookUpdate command.
- PhonebookFolderIDSearchResult
 - This is where the search results are saved when PhonebookFolderIDUpdate is called. The number of results displayed is set to 5 by default. Simply call SetText with the returned results to display the results on a button or label object.
- PhonebookFolderIDNavigation
 - This command is used for navigating through the Phonebook Folder Search Results. The navigation commands scroll through the results when there are more results than NumberofPhonebookFolderSearch variable.

Call History commands:

- CallHistoryRefresh
 - This is used to trigger the search for recent Call History based on the CallHistoryOccuranceType class variable.
- CallHistoryNavigation
 - This command is used for navigating through the CallHistory results.
- CallHistorySelect
 - When the CallHistorySelect is triggered, it will store the call history results' Callback Number into a class variable name, number. It can be accessed and use in the Dial String qualifier for the Hook Dial command.
- CallHistory
 - This is where the search results are saved when CallHistoryRefresh is called. The number of results displayed is set to 5 by default. Simply call SetText to display the results on a button or label object.

Control Commands, States, and Qualifiers:

ActiveSpeakerPIPPosition	'Center Left'	'Center Right'	'Lower Left'
	'Lower Right'	'Upper Center'	'Upper Left'
	'Upper Right'		
AudioInputLineChannel	'Left'	'Mono'	'Right'
['Input']	'1' – '4'		
AudioInputLineLevel	0 to 24 in steps of 1		
['Input']	'1' – '4'		
AudioInputLineMode	'On'	'Off'	
['Input']	'1' – '4'		
AudioInputLineSource	'1' – '5'		
['Input']	'1' – '4'		
AudioOutputLineChannel	'Left'	'Mono'	'Right'
['Output']	'1' – '6'		
AudioOutputLineLevel	-24 to 0 in steps of 1		
['Output']	'1' – '6'		
AudioOutputLineMode	'On'	'Off'	
['Output']	'1' – '6'		
AudioVolume	0 to 100 in steps of 1		
AudioVolumeMute	'On'	'Off'	
CallHistoryAcknowledgeAllMissedCalls	None		
CallHistoryAcknowledgeMissedCall	None		
['Missed Call Number']	1 to 2147483647		
CallHistoryDeleteAllEntries	'All'	'Missed'	'Placed'
	'Received'		
CallHistoryDeleteEntry	None		
['Entry Number']	1 to 2147483647 in steps of 1		
CallHistoryNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
CallHistoryRefresh	None		
CallHistorySelect	1 – 20		
CameraAssignedSerialNumberCommand	'String'		
['Camera ID']	'1' – '7'		

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Revision: 6/16/2017

CameraAutofocusCommand	'String'		
['Camera ID']	'1' – '7'		
CameraAutofocusMode	'Auto'	'Manual'	
['Camera ID']	'2' – '7'		
CameraFocus	'Far'	'Near'	'Stop'
['Camera ID']	'1' – '7'		
CameraPan	'Left'	'Right'	'Stop'
['Camera ID']	'1' – '7'		
['Pan Speed']	'1' – '15'		
CameraPreset	'Activate'	'Remove'	
['Preset Number']	'1' – '35'		
CameraPresetStore	'1' – '35'		
['Camera ID']	'1' – '7'		
['Default Position']	'False'	'True'	
CameraSpeakerTrack	'Activate'	'Deactivate'	
CameraSpeakerTrackTrackingMode	'Auto'	'Conservative'	
CameraTilt	'Down'	'Up'	'Stop'
['Camera ID']	'1' – '7'		
['Tilt Speed']	'1' – '15'		
CameraZoom	'In'	'Out'	'Stop'
['Camera ID']	'1' – '7'		
['Zoom Speed']	'1' – '15'		
ConferenceAutoAnswerDelay	0 to 50 in steps of 1		
ConferenceAutoAnswerMode	'On'	'Off'	
ConferenceAutoAnswerMute	'On'	'Off'	
ConferenceDefaultCallProtocol	'Auto'	'H323'	'Sip'
	'H320'		
ConferenceDefaultCallRate	64 to 6000 in steps of 1		
ConferenceDoNotDisturb	'On'	'Off'	
ConferenceEncryptionMode	'On'	'Off'	'Best Effort'
ConferenceFarEndControlMode	'On'	'Off'	
ConferenceSpeakerLock	'On'	'Off'	
['Speaker']	'Local'	'Remote'	
DNSDomainNameCommand	'String'		
DNSServerAddressCommand	'String'		

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['Server']	'1' – '3'		
DTMF	'0' – '9'	'*'	'#'
FarEndCameraPresetRecall	'1' – '15'		
FarEndCameraPTZ	'Left'	'Right'	'Up'
	'Down'	'Zoom In'	'Zoom Out'
	'Stop'		
FarEndCameraSourceSelect	'0' – '15'		
H323AliasE164Command	'String'		
H323AliasIDCommand	'String'		
H323AuthenticationMode	'On'	'Off'	
H323CallSetupMode	'Direct'	'Gatekeeper'	
H323GatekeeperAddressCommand	'String'		
H323LoginNameCommand	'String'		
H323PasswordCommand	'String'		
Hook	'Dial'	'Join 1'	'Accept'
	'Reject'	'Hold 1'	'Resume 1'
	'Disconnect 1'	'Hold 2'	'Hold 3'
	'Resume 2'	'Resume 3'	'Disconnect 2'
	'Disconnect 3'	'Join 2'	'Join 3'
	'Join 4'	'Join 5'	'Hold 4'
	'Hold 5'	'Disconnect 4'	'Disconnect 5'
	'Resume 4'	'Resume 5'	
['Protocol']	'H320'	'H323'	'SIP'
	'Auto'		
['Number'] ¹	'String'	None	
IPv4AddressCommand	'String'		
IPv4GatewayCommand	'String'		
IPv4SubnetMaskCommand	'String'		
MicrophoneLevel	0 to 70 in steps of 1		
['Microphone']	'1' – '8'		
MicrophoneMode	'On'	'Off'	
['Microphone']	'1' – '8'		
MuteAllMicrophones	'On'	'Off'	
PhonebookFolderIDNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
['Phonebook Type']	'Corporate'	'Local'	

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PhonebookFolderIDSearchSet	1 – 10		
PhonebookFolderIDUpdate	None		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookSearch	'String'		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookSearchSet	1 – 10		
PhonebookUpdate	'Refresh'	'Next Search'	'Previous Search'
['Phonebook Type']	'Corporate'	'Local'	
['Contact']	'String'		
['Folder ID']	'String'		
PresentationPIPPosition	'Center Left'	'Center Right'	'Lower Left'
	'Lower Right'	'Upper Center'	'Upper Left'
	'Upper Right'		
SelfView	'On'	'Off'	
SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left'	'Upper Center'	'Upper Right'
	'Center Left'	'Center Right'	'Lower Left'
	'Lower Right'		
Standby	'On'	'Off'	
VideoInput	'1' – '4'		
['Connector ID']	'1' – '5'		

¹ Number qualifier is needed when the 'Dial' state is selected for Hook command, otherwise it can be set to None.

Global Scriptor Module Communication Sheet

Status Available:

ActiveSpeakerPIPPosition	'Center Left' 'Lower Right' 'Upper Right'	'Center Right' 'Upper Center'	'Lower Left' 'Upper Left'
AudioInputLineLevel	0 to 24 in steps of 1		
['Input']	'1' – '4'		
AudioOutputLineLevel	-24 to 0 in steps of 1		
['Output']	'1' – '6'		
AudioVolume	0 to 100 in steps of 1		
AudioVolumeMute	'On' 'Off'		
CallAnswerStatus	'Unanswered' 'Answered'	'Ignored'	'Auto Answered'
['Call']	'1' – '5'		
CallHistory	'String'		
['Button']	'1' – '20'		
['Detail Type']	'Display Name' 'Occurrence Count'	'Callback Number' 'Occurrence Type'	'Last Occurrence Time'
CallID	None		
CallOnHold	'Held' 'Not Held'		
['Call']	'1' – '5'		
CallProtocol	'H320'	'H323'	'Sip'
['Call']	'1' – '5'		
CallStatus	'Idle' 'Connecting' 'On Hold' 'Remote Preserved'	'Dialing' 'Connected' 'Early Media'	'Ringing' 'Disconnecting' 'Preserved'
['Call']	'1' – '5'		
CallStatusType	'Video' 'Forward All Calls'	'Audio' 'Unknown'	'Audio Can Escalate'
['Call']	'1' – '5'		
ConferenceAutoAnswerDelay	0 to 50 in steps of 1		
ConferenceDefaultCallRate	64 to 6000 in steps of 1		
ConferenceDoNotDisturb	'On' 'Off'		

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ConferenceFarEndControlMode	'On'	'Off'	
ConferenceSpeakerLock	'On'	'Off'	
['Speaker']	'Local'	'Remote'	
ConnectionStatus	'Connected'	'Disconnected'	
DisplayName	'String'		
['Call']	'1' – '5'		
DNSServerAddressStatus	'String'		
['Server']	'1' – '3'		
H323GatekeeperAddressStatus	'String'		
H323GatekeeperStatus	'Required'	'Discovering'	'Discovered'
	'Authenticating'	'Authenticated'	'Registering'
	'Registered'	'Inactive'	'Rejected'
IPv4AddressStatus	'String'		
IPv4GatewayStatus	'String'		
IPv4SubnetMaskStatus	'String'		
MicrophoneLevel	0 to 70 in steps of 1		
['Microphone']	'1' – '8'		
MuteAllMicrophones	'On'	'Off'	
PhonebookFolderID	'String'		
PhonebookFolderIDSearchResult	'String'		
['Button']	1 to 10 in steps of 1		
PhonebookSearch	'String'		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookSearchResult	'String'		
['Button']	1 to 10 in steps of 1		
PresentationMode	'Off'	'Sending'	'Receiving'
RemoteNumber	'String'		
['Call']	'1' – '5'		
SelfView	'On'	'Off'	
SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left'	'Upper Center'	'Upper Right'
	'Center Left'	'Center Right'	'Lower Left'
	'Lower Right'		
Standby	'On'	'Off'	
VideoInput	'1' – '4'		

['Connector ID']	'1' - '5'
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Cable and Adapter Requirements:

Captive Screw to Male DB9 RS-232 Serial Cable

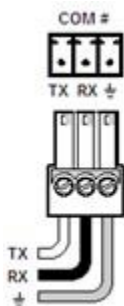
Notes for the Device:

Serial communication:

Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type: Ethernet

Default Port: 23

Logon Credentials Supported: Yes

Default User Name: admin

Multi-Connection Capabilities: Yes

Port Changeability: No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Telnet mode must be set to 'On' by an admin in order to have open communication with the driver

Appendix A. Set Commands

Active Speaker PIP Position Center Left	xCommand Video ActiveSpeakerPIP Set Position: CenterLeft\x0D
Active Speaker PIP Position Center Right	xCommand Video ActiveSpeakerPIP Set Position: CenterRight\x0D
Active Speaker PIP Position Lower Left	xCommand Video ActiveSpeakerPIP Set Position: LowerLeft\x0D
Active Speaker PIP Position Lower Right	xCommand Video ActiveSpeakerPIP Set Position: LowerRight\x0D
Active Speaker PIP Position Upper Center	xCommand Video ActiveSpeakerPIP Set Position: UpperCenter\x0D
Active Speaker PIP Position Upper Left	xCommand Video ActiveSpeakerPIP Set Position: UpperLeft\x0D
Active Speaker PIP Position Upper Right	xCommand Video ActiveSpeakerPIP Set Position: UpperRight\x0D
Audio Input Line Channel Left Input 1	xConfiguration Audio Input Line 1 Channel: Left\x0D
Audio Input Line Channel Left Input 4	xConfiguration Audio Input Line 4 Channel: Left\x0D
Audio Input Line Channel Mono Input 1	xConfiguration Audio Input Line 1 Channel: Mono\x0D
Audio Input Line Channel Mono Input 4	xConfiguration Audio Input Line 4 Channel: Mono\x0D
Audio Input Line Channel Right Input 1	xConfiguration Audio Input Line 1 Channel: Right\x0D
Audio Input Line Channel Right Input 4	xConfiguration Audio Input Line 4 Channel: Right\x0D
Audio Input Line Level 0 Input 1	xConfiguration Audio Input Line 1 Level: 0\x0D
Audio Input Line Level 0 Input 4	xConfiguration Audio Input Line 4 Level: 0\x0D
Audio Input Line Level 24 Input 1	xConfiguration Audio Input Line 1 Level: 24\x0D
Audio Input Line Level 24 Input 4	xConfiguration Audio Input Line 4 Level: 24\x0D
Audio Input Line Mode Off Input 1	xConfiguration Audio Input Line 1 Mode: Off\x0D
Audio Input Line Mode Off Input 4	xConfiguration Audio Input Line 4 Mode: Off\x0D
Audio Input Line Mode On Input 1	xConfiguration Audio Input Line 1 Mode: On\x0D
Audio Input Line Mode On Input 4	xConfiguration Audio Input Line 4 Mode: On\x0D
Audio Input Line Source 1 Input 1	xConfiguration Audio Input Line 1 VideoAssociation VideoInputSource: 1\x0D
Audio Input Line Source 1 Input 4	xConfiguration Audio Input Line 4 VideoAssociation VideoInputSource: 1\x0D
Audio Input Line Source 5 Input 1	xConfiguration Audio Input Line 1 VideoAssociation VideoInputSource: 5\x0D
Audio Input Line Source 5 Input 4	xConfiguration Audio Input Line 4 VideoAssociation VideoInputSource: 5\x0D
Audio Output Line Channel Left Output 1	xConfiguration Audio Output Line 1 Channel: Left\x0D
Audio Output Line Channel Left Output 6	xConfiguration Audio Output Line 6 Channel: Left\x0D
Audio Output Line Channel Mono Output 1	xConfiguration Audio Output Line 1 Channel: Mono\x0D
Audio Output Line Channel Mono Output 6	xConfiguration Audio Output Line 6 Channel: Mono\x0D
Audio Output Line Channel Right Output 1	xConfiguration Audio Output Line 1 Channel: Right\x0D
Audio Output Line Channel Right Output 6	xConfiguration Audio Output Line 6 Channel: Right\x0D
Audio Output Line Level 0 Output 1	xConfiguration Audio Output Line 1 Level: 0\x0D

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Audio Output Line Level 0 Output 6	xConfiguration Audio Output Line 6 Level: 0\x0D
Audio Output Line Level -24 Output 1	xConfiguration Audio Output Line 1 Level: -24\x0D
Audio Output Line Level -24 Output 6	xConfiguration Audio Output Line 6 Level: -24\x0D
Audio Output Line Mode Off Output 1	xConfiguration Audio Output Line 1 Mode: Off\x0D
Audio Output Line Mode Off Output 6	xConfiguration Audio Output Line 6 Mode: Off\x0D
Audio Output Line Mode On Output 1	xConfiguration Audio Output Line 1 Mode: On\x0D
Audio Output Line Mode On Output 6	xConfiguration Audio Output Line 6 Mode: On\x0D
Audio Volume 0	xCommand Audio Volume Set Level: 0\x0D
Audio Volume 100	xCommand Audio Volume Set Level: 100\x0D
Audio Volume Mute Off	xCommand Audio Volume Unmute\x0D
Audio Volume Mute On	xCommand Audio Volume Mute\x0D
Call History Acknowledge All Missed Calls None	xCommand CallHistory AcknowledgeAllMissedCalls\x0D
Call History Acknowledge Missed Call None Missed Call Number 1	xCommand CallHistory AcknowledgeMissedCall CallHistoryID: 1\x0D
Call History Acknowledge Missed Call None Missed Call Number 2147483647	xCommand CallHistory AcknowledgeMissedCall CallHistoryID: 2147483647\x0D
Call History Delete All Entries All	xCommand CallHistory DeleteAll Filter: All\x0D
Call History Delete All Entries Missed	xCommand CallHistory DeleteAll Filter: Missed\x0D
Call History Delete All Entries Placed	xCommand CallHistory DeleteAll Filter: Placed\x0D
Call History Delete All Entries Received	xCommand CallHistory DeleteAll Filter: Received\x0D
Call History Delete Entry None Entry Number 1	xCommand CallHistory DeleteEntry CallHistoryID: 1\x0D
Call History Delete Entry None Entry Number 2147483647	xCommand CallHistory DeleteEntry CallHistoryID: 2147483647\x0D
Camera Autofocus Command None Camera ID 1	xCommand Camera TriggerAutofocus CameraId: 1\x0D
Camera Autofocus Command None Camera ID 7	xCommand Camera TriggerAutofocus CameraId: 7\x0D
Camera Autofocus Mode Auto Camera ID 2	xConfiguration Cameras Camera 2 Focus Mode: Auto\x0D
Camera Autofocus Mode Auto Camera ID 7	xConfiguration Cameras Camera 7 Focus Mode: Auto\x0D
Camera Autofocus Mode Manual Camera ID 2	xConfiguration Cameras Camera 2 Focus Mode: Manual\x0D
Camera Autofocus Mode Manual Camera ID 7	xConfiguration Cameras Camera 7 Focus Mode: Manual\x0D
Camera Focus Far Camera ID 1	xCommand Camera Ramp CameraId: 1 Focus: Far\x0D
Camera Focus Far Camera ID 7	xCommand Camera Ramp CameraId: 7 Focus: Far\x0D
Camera Focus Near Camera ID 1	xCommand Camera Ramp CameraId: 1 Focus: Near\x0D
Camera Focus Near Camera ID 7	xCommand Camera Ramp CameraId: 7 Focus: Near\x0D
Camera Focus Stop Camera ID 1	xCommand Camera Ramp CameraId: 1 Focus: Stop\x0D
Camera Focus Stop Camera ID 7	xCommand Camera Ramp CameraId: 7 Focus: Stop\x0D
Camera Pan Left Camera ID 1 Pan Speed 1	xCommand Camera Ramp CameraId: 1 PanSpeed: 1 Pan: Left\x0D

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Camera Pan Left Camera ID 1 Pan Speed 15	xCommand Camera Ramp CameraId: 1 PanSpeed: 15 Pan: Left\x0D
Camera Pan Left Camera ID 7 Pan Speed 1	xCommand Camera Ramp CameraId: 7 PanSpeed: 1 Pan: Left\x0D
Camera Pan Left Camera ID 7 Pan Speed 15	xCommand Camera Ramp CameraId: 7 PanSpeed: 15 Pan: Left\x0D
Camera Pan Right Camera ID 1 Pan Speed 1	xCommand Camera Ramp CameraId: 1 PanSpeed: 1 Pan: Right\x0D
Camera Pan Right Camera ID 1 Pan Speed 15	xCommand Camera Ramp CameraId: 1 PanSpeed: 15 Pan: Right\x0D
Camera Pan Right Camera ID 7 Pan Speed 1	xCommand Camera Ramp CameraId: 7 PanSpeed: 1 Pan: Right\x0D
Camera Pan Right Camera ID 7 Pan Speed 15	xCommand Camera Ramp CameraId: 7 PanSpeed: 15 Pan: Right\x0D
Camera Pan Stop Camera ID 1 Pan Speed 1	xCommand Camera Ramp CameraId: 1 PanSpeed: 1 Pan: Stop\x0D
Camera Pan Stop Camera ID 1 Pan Speed 15	xCommand Camera Ramp CameraId: 1 PanSpeed: 15 Pan: Stop\x0D
Camera Pan Stop Camera ID 7 Pan Speed 1	xCommand Camera Ramp CameraId: 7 PanSpeed: 1 Pan: Stop\x0D
Camera Pan Stop Camera ID 7 Pan Speed 15	xCommand Camera Ramp CameraId: 7 PanSpeed: 15 Pan: Stop\x0D
Camera Preset Activate Preset Number 1	xCommand Camera Preset 1 PresetID: Activate\x0D
Camera Preset Activate Preset Number 35	xCommand Camera Preset 35 PresetID: Activate\x0D
Camera Preset Remove Preset Number 1	xCommand Camera Preset 1 PresetID: Remove\x0D
Camera Preset Remove Preset Number 35	xCommand Camera Preset 35 PresetID: Remove\x0D
Camera Preset Store 1 Camera ID 1 Default Position False	xCommand Camera Preset Store PresetId: 1 CameraId: 1 DefaultPosition: False\x0D
Camera Preset Store 1 Camera ID 1 Default Position True	xCommand Camera Preset Store PresetId: 1 CameraId: 1 DefaultPosition: True\x0D
Camera Preset Store 1 Camera ID 7 Default Position False	xCommand Camera Preset Store PresetId: 1 CameraId: 7 DefaultPosition: False\x0D
Camera Preset Store 1 Camera ID 7 Default Position True	xCommand Camera Preset Store PresetId: 1 CameraId: 7 DefaultPosition: True\x0D
Camera Preset Store 35 Camera ID 1 Default Position False	xCommand Camera Preset Store PresetId: 35 CameraId: 1 DefaultPosition: False\x0D
Camera Preset Store 35 Camera ID 1 Default Position True	xCommand Camera Preset Store PresetId: 35 CameraId: 1 DefaultPosition: True\x0D
Camera Preset Store 35 Camera ID 7 Default Position False	xCommand Camera Preset Store PresetId: 35 CameraId: 7 DefaultPosition: False\x0D
Camera Preset Store 35 Camera ID 7 Default Position True	xCommand Camera Preset Store PresetId: 35 CameraId: 7 DefaultPosition: True\x0D
Camera Speaker Track Activate	xCommand Cameras SpeakerTrack Activate\x0D
Camera Speaker Track Deactivate	xCommand Cameras SpeakerTrack Deactivate\x0D
Camera Speaker Track Tracking Mode Auto	xCommand Cameras SpeakerTrack TrackingMode: Auto\x0D
Camera Speaker Track Tracking Mode Conservative	xCommand Cameras SpeakerTrack TrackingMode: Conservative\x0D

Camera Tilt Down Camera ID 1 Tilt Speed 1	xCommand Camera Ramp CameraId: 1 Tilt: Down TiltSpeed: 1\x0D
Camera Tilt Down Camera ID 1 Tilt Speed 15	xCommand Camera Ramp CameraId: 1 Tilt: Down TiltSpeed: 15\x0D
Camera Tilt Down Camera ID 7 Tilt Speed 1	xCommand Camera Ramp CameraId: 7 Tilt: Down TiltSpeed: 1\x0D
Camera Tilt Down Camera ID 7 Tilt Speed 15	xCommand Camera Ramp CameraId: 7 Tilt: Down TiltSpeed: 15\x0D
Camera Tilt Stop Camera ID 1 Tilt Speed 1	xCommand Camera Ramp CameraId: 1 Tilt: Stop TiltSpeed: 1\x0D
Camera Tilt Stop Camera ID 1 Tilt Speed 15	xCommand Camera Ramp CameraId: 1 Tilt: Stop TiltSpeed: 15\x0D
Camera Tilt Stop Camera ID 7 Tilt Speed 1	xCommand Camera Ramp CameraId: 7 Tilt: Stop TiltSpeed: 1\x0D
Camera Tilt Stop Camera ID 7 Tilt Speed 15	xCommand Camera Ramp CameraId: 7 Tilt: Stop TiltSpeed: 15\x0D
Camera Tilt Up Camera ID 1 Tilt Speed 1	xCommand Camera Ramp CameraId: 1 Tilt: Up TiltSpeed: 1\x0D
Camera Tilt Up Camera ID 1 Tilt Speed 15	xCommand Camera Ramp CameraId: 1 Tilt: Up TiltSpeed: 15\x0D
Camera Tilt Up Camera ID 7 Tilt Speed 1	xCommand Camera Ramp CameraId: 7 Tilt: Up TiltSpeed: 1\x0D
Camera Tilt Up Camera ID 7 Tilt Speed 15	xCommand Camera Ramp CameraId: 7 Tilt: Up TiltSpeed: 15\x0D
Camera Zoom In Camera ID 1 Zoom Speed 1	xCommand Camera Ramp CameraId: 1 Zoom: In ZoomSpeed: 1\x0D
Camera Zoom In Camera ID 1 Zoom Speed 15	xCommand Camera Ramp CameraId: 1 Zoom: In ZoomSpeed: 15\x0D
Camera Zoom In Camera ID 7 Zoom Speed 1	xCommand Camera Ramp CameraId: 7 Zoom: In ZoomSpeed: 1\x0D
Camera Zoom In Camera ID 7 Zoom Speed 15	xCommand Camera Ramp CameraId: 7 Zoom: In ZoomSpeed: 15\x0D
Camera Zoom Out Camera ID 1 Zoom Speed 1	xCommand Camera Ramp CameraId: 1 Zoom: Out ZoomSpeed: 1\x0D
Camera Zoom Out Camera ID 1 Zoom Speed 15	xCommand Camera Ramp CameraId: 1 Zoom: Out ZoomSpeed: 15\x0D
Camera Zoom Out Camera ID 7 Zoom Speed 1	xCommand Camera Ramp CameraId: 7 Zoom: Out ZoomSpeed: 1\x0D
Camera Zoom Out Camera ID 7 Zoom Speed 15	xCommand Camera Ramp CameraId: 7 Zoom: Out ZoomSpeed: 15\x0D
Camera Zoom Stop Camera ID 1 Zoom Speed 1	xCommand Camera Ramp CameraId: 1 Zoom: Stop ZoomSpeed: 1\x0D
Camera Zoom Stop Camera ID 1 Zoom Speed 15	xCommand Camera Ramp CameraId: 1 Zoom: Stop ZoomSpeed: 15\x0D
Camera Zoom Stop Camera ID 7 Zoom Speed 1	xCommand Camera Ramp CameraId: 7 Zoom: Stop ZoomSpeed: 1\x0D
Camera Zoom Stop Camera ID 7 Zoom Speed 15	xCommand Camera Ramp CameraId: 7 Zoom: Stop ZoomSpeed: 15\x0D
Conference AutoAnswer Delay 0	xConfiguration Conference AutoAnswer Delay: 0\x0D

Conference AutoAnswer Delay 50	xConfiguration Conference AutoAnswer Delay: 50\x0D
Conference AutoAnswer Mode Off	xConfiguration Conference AutoAnswer Mode: Off\x0D
Conference AutoAnswer Mode On	xConfiguration Conference AutoAnswer Mode: On\x0D
Conference AutoAnswer Mute Off	xConfiguration Conference AutoAnswer Mute: Off\x0D
Conference AutoAnswer Mute On	xConfiguration Conference AutoAnswer Mute: On\x0D
Conference Default Call Protocol Auto	xConfiguration Conference DefaultCall Protocol: Auto\x0D
Conference Default Call Protocol H320	xConfiguration Conference DefaultCall Protocol: H320\x0D
Conference Default Call Protocol H323	xConfiguration Conference DefaultCall Protocol: H323\x0D
Conference Default Call Protocol Sip	xConfiguration Conference DefaultCall Protocol: Sip\x0D
Conference Default Call Rate 6000	xConfiguration Conference DefaultCall Rate: 6000\x0D
Conference Default Call Rate 64	xConfiguration Conference DefaultCall Rate: 64\x0D
Conference Do Not Disturb Off	xCommand Conference DoNotDisturb Deactivate\x0D
Conference Do Not Disturb On	xCommand Conference DoNotDisturb Activate\x0D
Conference Encryption Mode Best Effort	xConfiguration Conference Encryption Mode: BestEffort\x0D
Conference Encryption Mode Off	xConfiguration Conference Encryption Mode: Off\x0D
Conference Encryption Mode On	xConfiguration Conference Encryption Mode: On\x0D
Conference Far End Control Mode Off	xConfiguration Conference FarEndControl Mode: Off\x0D
Conference Far End Control Mode On	xConfiguration Conference FarEndControl Mode: On\x0D
Conference Speaker Lock Off Speaker Local	xCommand Conference SpeakerLock Release\x0D
Conference Speaker Lock Off Speaker Remote	xCommand Conference SpeakerLock Release\x0D
Conference Speaker Lock On Speaker Local	xCommand Conference SpeakerLock Set Target: local\x0D
Conference Speaker Lock On Speaker Remote	xCommand Conference SpeakerLock Set Target: remote\x0D
DTMF #	xCommand Call DTMFSend DTMFString: "#"\x0D
DTMF *	xCommand Call DTMFSend DTMFString: "*"\x0D
DTMF 0	xCommand Call DTMFSend DTMFString: "0"\x0D
DTMF 9	xCommand Call DTMFSend DTMFString: "9"\x0D
Far End Camera Preset Recall 1	xCommand FarEndControl RoomPreset Activate PresetId: 1\x0D
Far End Camera Preset Recall 15	xCommand FarEndControl RoomPreset Activate PresetId: 15\x0D
Far End Camera PTZ Down	xCommand FarEndControl Camera Move Value: Down\x0D
Far End Camera PTZ Left	xCommand FarEndControl Camera Move Value: Left\x0D
Far End Camera PTZ Right	xCommand FarEndControl Camera Move Value: Right\x0D
Far End Camera PTZ Stop	xCommand FarEndControl Camera Stop\x0D

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Far End Camera PTZ Up	xCommand FarEndControl Camera Move Value: Up\x0D
Far End Camera PTZ Zoom In	xCommand FarEndControl Camera Move Value: ZoomIn\x0D
Far End Camera PTZ Zoom Out	xCommand FarEndControl Camera Move Value: ZoomOut\x0D
Far End Camera Source Select 0	xCommand FarEndControl Source Select SourceId: 0\x0D
Far End Camera Source Select 15	xCommand FarEndControl Source Select SourceId: 15\x0D
H323 Authentication Mode Off	xConfiguration H323 Authentication Mode: Off\x0D
H323 Authentication Mode On	xConfiguration H323 Authentication Mode: On\x0D
H323 Call Setup Mode Direct	xConfiguration H323 CallSetup Mode: Direct\x0D
H323 Call Setup Mode Gatekeeper	xConfiguration H323 CallSetup Mode: Gatekeeper\x0D
Hook Dial Protocol Auto Number value	xCommand Dial Number:"value"\x0D
Hook Dial Protocol H320 Number value	xCommand Dial Number:"value" Protocol:h320\x0D
Hook Dial Protocol H323 Number value	xCommand Dial Number:"value" Protocol:h323\x0D
Hook Dial Protocol SIP Number value	xCommand Dial Number:"value" Protocol:sip\x0D
Hook Join 1 Protocol Auto Number value	xCommand Call Join CallId: (call id of call 1)\x0D
Hook Join 1 Protocol H320 Number value	xCommand Call Join CallId: (call id of call 1)\x0D
Hook Join 1 Protocol H323 Number value	xCommand Call Join CallId: (call id of call 1)\x0D
Hook Join 1 Protocol SIP Number value	xCommand Call Join CallId: (call id of call 1)\x0D
Hook Join 2 Protocol Auto Number value	xCommand Call Join CallId: (call id of call 2)\x0D
Hook Join 2 Protocol H320 Number value	xCommand Call Join CallId: (call id of call 2)\x0D
Hook Join 2 Protocol H323 Number value	xCommand Call Join CallId: (call id of call 2)\x0D
Hook Join 2 Protocol SIP Number value	xCommand Call Join CallId: (call id of call 2)\x0D
Hook Join 3 Protocol Auto Number value	xCommand Call Join CallId: (call id of call 3)\x0D
Hook Join 3 Protocol H320 Number value	xCommand Call Join CallId: (call id of call 3)\x0D
Hook Join 3 Protocol H323 Number value	xCommand Call Join CallId: (call id of call 3)\x0D
Hook Join 3 Protocol SIP Number value	xCommand Call Join CallId: (call id of call 3)\x0D
Hook Join 4 Protocol Auto Number value	xCommand Call Join CallId: (call id of call 4)\x0D
Hook Join 4 Protocol H320 Number value	xCommand Call Join CallId: (call id of call 4)\x0D
Hook Join 4 Protocol H323 Number value	xCommand Call Join CallId: (call id of call 4)\x0D
Hook Join 4 Protocol SIP Number value	xCommand Call Join CallId: (call id of call 4)\x0D
Hook Join 5 Protocol Auto Number value	xCommand Call Join CallId: (call id of call 5)\x0D
Hook Join 5 Protocol H320 Number value	xCommand Call Join CallId: (call id of call 5)\x0D
Hook Join 5 Protocol H323 Number value	xCommand Call Join CallId: (call id of call 5)\x0D
Hook Join 5 Protocol SIP Number value	xCommand Call Join CallId: (call id of call 5)\x0D
Hook Resume 1 Protocol Auto Number value	xCommand Call Resume CallId: (call id of call 1)\x0D
Hook Resume 1 Protocol H320 Number value	xCommand Call Resume CallId: (call id of call 1)\x0D
Hook Resume 1 Protocol H323 Number value	xCommand Call Resume CallId: (call id of call 1)\x0D
Hook Resume 1 Protocol SIP Number value	xCommand Call Resume CallId: (call id of call 1)\x0D
Hook Resume 2 Protocol Auto Number value	xCommand Call Resume CallId: (call id of call 2)\x0D

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Hook Resume 2 Protocol H320 Number value	xCommand Call Resume CallId: (call id of call 2)\x0D
Hook Resume 2 Protocol H323 Number value	xCommand Call Resume CallId: (call id of call 2)\x0D
Hook Resume 2 Protocol SIP Number value	xCommand Call Resume CallId: (call id of call 2)\x0D
Hook Resume 3 Protocol Auto Number value	xCommand Call Resume CallId: (call id of call 3)\x0D
Hook Resume 3 Protocol H320 Number value	xCommand Call Resume CallId: (call id of call 3)\x0D
Hook Resume 3 Protocol H323 Number value	xCommand Call Resume CallId: (call id of call 3)\x0D
Hook Resume 3 Protocol SIP Number value	xCommand Call Resume CallId: (call id of call 3)\x0D
Hook Resume 4 Protocol Auto Number value	xCommand Call Resume CallId: (call id of call 4)\x0D
Hook Resume 4 Protocol H320 Number value	xCommand Call Resume CallId: (call id of call 4)\x0D
Hook Resume 4 Protocol H323 Number value	xCommand Call Resume CallId: (call id of call 4)\x0D
Hook Resume 4 Protocol SIP Number value	xCommand Call Resume CallId: (call id of call 4)\x0D
Hook Resume 5 Protocol Auto Number value	xCommand Call Resume CallId: (call id of call 5)\x0D
Hook Resume 5 Protocol H320 Number value	xCommand Call Resume CallId: (call id of call 5)\x0D
Hook Resume 5 Protocol H323 Number value	xCommand Call Resume CallId: (call id of call 5)\x0D
Hook Resume 5 Protocol SIP Number value	xCommand Call Resume CallId: (call id of call 5)\x0D
Hook Hold 1 Protocol Auto Number value	xCommand Call Hold CallId: (call id of call 1)\x0D
Hook Hold 1 Protocol H320 Number value	xCommand Call Hold CallId: (call id of call 1)\x0D
Hook Hold 1 Protocol H323 Number value	xCommand Call Hold CallId: (call id of call 1)\x0D
Hook Hold 1 Protocol SIP Number value	xCommand Call Hold CallId: (call id of call 1)\x0D
Hook Hold 2 Protocol Auto Number value	xCommand Call Hold CallId: (call id of call 2)\x0D
Hook Hold 2 Protocol H320 Number value	xCommand Call Hold CallId: (call id of call 2)\x0D
Hook Hold 2 Protocol H323 Number value	xCommand Call Hold CallId: (call id of call 2)\x0D
Hook Hold 2 Protocol SIP Number value	xCommand Call Hold CallId: (call id of call 2)\x0D
Hook Hold 3 Protocol Auto Number value	xCommand Call Hold CallId: (call id of call 3)\x0D
Hook Hold 3 Protocol H320 Number value	xCommand Call Hold CallId: (call id of call 3)\x0D
Hook Hold 3 Protocol H323 Number value	xCommand Call Hold CallId: (call id of call 3)\x0D
Hook Hold 3 Protocol SIP Number value	xCommand Call Hold CallId: (call id of call 3)\x0D
Hook Hold 4 Protocol Auto Number value	xCommand Call Hold CallId: (call id of call 4)\x0D

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Hook Hold 4 Protocol H320 Number value	xCommand Call Hold CallId: (call id of call 4)\x0D
Hook Hold 4 Protocol H323 Number value	xCommand Call Hold CallId: (call id of call 4)\x0D
Hook Hold 4 Protocol SIP Number value	xCommand Call Hold CallId: (call id of call 4)\x0D
Hook Hold 5 Protocol Auto Number value	xCommand Call Hold CallId: (call id of call 5)\x0D
Hook Hold 5 Protocol H320 Number value	xCommand Call Hold CallId: (call id of call 5)\x0D
Hook Hold 5 Protocol H323 Number value	xCommand Call Hold CallId: (call id of call 5)\x0D
Hook Hold 5 Protocol SIP Number value	xCommand Call Hold CallId: (call id of call 5)\x0D
Hook Disconnect 1 Protocol Auto Number value	xCommand Call Disconnect CallId: (call id of call 1)\x0D
Hook Disconnect 1 Protocol H320 Number value	xCommand Call Disconnect CallId: (call id of call 1)\x0D
Hook Disconnect 1 Protocol H323 Number value	xCommand Call Disconnect CallId: (call id of call 1)\x0D
Hook Disconnect 1 Protocol SIP Number value	xCommand Call Disconnect CallId: (call id of call 1)\x0D
Hook Disconnect 2 Protocol Auto Number value	xCommand Call Disconnect CallId: (call id of call 2)\x0D
Hook Disconnect 2 Protocol H320 Number value	xCommand Call Disconnect CallId: (call id of call 2)\x0D
Hook Disconnect 2 Protocol H323 Number value	xCommand Call Disconnect CallId: (call id of call 2)\x0D
Hook Disconnect 2 Protocol SIP Number value	xCommand Call Disconnect CallId: (call id of call 2)\x0D
Hook Disconnect 3 Protocol Auto Number value	xCommand Call Disconnect CallId: (call id of call 3)\x0D
Hook Disconnect 3 Protocol H320 Number value	xCommand Call Disconnect CallId: (call id of call 3)\x0D
Hook Disconnect 3 Protocol H323 Number value	xCommand Call Disconnect CallId: (call id of call 3)\x0D
Hook Disconnect 3 Protocol SIP Number value	xCommand Call Disconnect CallId: (call id of call 3)\x0D
Hook Disconnect 4 Protocol Auto Number value	xCommand Call Disconnect CallId: (call id of call 4)\x0D
Hook Disconnect 4 Protocol H320 Number value	xCommand Call Disconnect CallId: (call id of call 4)\x0D
Hook Disconnect 4 Protocol H323 Number value	xCommand Call Disconnect CallId: (call id of call 4)\x0D
Hook Disconnect 4 Protocol SIP Number value	xCommand Call Disconnect CallId: (call id of call 4)\x0D
Hook Disconnect 5 Protocol Auto Number value	xCommand Call Disconnect CallId: (call id of call 5)\x0D
Hook Disconnect 5 Protocol H320 Number value	xCommand Call Disconnect CallId: (call id of call 5)\x0D
Hook Disconnect 5 Protocol H323 Number value	xCommand Call Disconnect CallId: (call id of call 5)\x0D

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Hook Disconnect 5 Protocol SIP Number value	xCommand Call Disconnect CallId: (call id of call 5)\x0D
Hook Accept Protocol Auto Number None	xCommand Call Accept\x0D
Hook Accept Protocol H320 Number None	xCommand Call Accept\x0D
Hook Accept Protocol H323 Number None	xCommand Call Accept\x0D
Hook Accept Protocol SIP Number None	xCommand Call Accept\x0D
Hook Reject Protocol Auto Number None	xCommand Call Reject\x0D
Hook Reject Protocol H320 Number None	xCommand Call Reject\x0D
Hook Reject Protocol H323 Number None	xCommand Call Reject\x0D
Hook Reject Protocol SIP Number None	xCommand Call Reject\x0D
Microphone Level 0 Microphone 1	xConfiguration Audio Input Microphone 1 Level: 0\x0D
Microphone Level 0 Microphone 8	xConfiguration Audio Input Microphone 8 Level: 0\x0D
Microphone Level 70 Microphone 1	xConfiguration Audio Input Microphone 1 Level: 70\x0D
Microphone Level 70 Microphone 8	xConfiguration Audio Input Microphone 8 Level: 70\x0D
Microphone Mode Off Microphone 1	xConfiguration Audio Input Microphone 1 Mode: Off\x0D
Microphone Mode Off Microphone 8	xConfiguration Audio Input Microphone 8 Mode: Off\x0D
Microphone Mode On Microphone 1	xConfiguration Audio Input Microphone 1 Mode: On\x0D
Microphone Mode On Microphone 8	xConfiguration Audio Input Microphone 8 Mode: On\x0D
Mute All Microphones Off	xCommand Microphones Unmute\x0D
Mute All Microphones On	xCommand Microphones Mute\x0D
Presentation PIP Position Center Left	xCommand Video PresentationPIP Set Position: CenterLeft\x0D
Presentation PIP Position Center Right	xCommand Video PresentationPIP Set Position: CenterRight\x0D
Presentation PIP Position Lower Left	xCommand Video PresentationPIP Set Position: LowerLeft\x0D
Presentation PIP Position Lower Right	xCommand Video PresentationPIP Set Position: LowerRight\x0D
Presentation PIP Position Upper Center	xCommand Video PresentationPIP Set Position: UpperCenter\x0D
Presentation PIP Position Upper Left	xCommand Video PresentationPIP Set Position: UpperLeft\x0D
Presentation PIP Position Upper Right	xCommand Video PresentationPIP Set Position: UpperRight\x0D
Self View Default Fullscreen Mode Current	xConfiguration Video Selfview Default FullscreenMode: Current\x0D
Self View Default Fullscreen Mode Off	xConfiguration Video Selfview Default FullscreenMode: Off\x0D
Self View Default Fullscreen Mode On	xConfiguration Video Selfview Default FullscreenMode: On\x0D
Self View Off	xCommand Video Selfview Set Mode: Off\x0D

Self View On	xCommand Video Selfview Set Mode: On\x0D
Self View Position Center Left	xConfiguration Video Selfview Default PIPPosition: CenterLeft\x0D
Self View Position Center Right	xConfiguration Video Selfview Default PIPPosition: CenterRight\x0D
Self View Position Lower Left	xConfiguration Video Selfview Default PIPPosition: LowerLeft\x0D
Self View Position Lower Right	xConfiguration Video Selfview Default PIPPosition: LowerRight\x0D
Self View Position Upper Center	xConfiguration Video Selfview Default PIPPosition: UpperCenter\x0D
Self View Position Upper Left	xConfiguration Video Selfview Default PIPPosition: UpperLeft\x0D
Self View Position Upper Right	xConfiguration Video Selfview Default PIPPosition: UpperRight\x0D
Standby Off	xCommand Standby Deactivate\x0D
Standby On	xCommand Standby Activate\x0D
Video Input 1 Connector ID 1	xCommand Video Input SetMainVideoSource ConnectorID: 1 SourceId: 1\x0D
Video Input 1 Connector ID 5	xCommand Video Input SetMainVideoSource ConnectorID: 5 SourceId: 1\x0D
Video Input 4 Connector ID 1	xCommand Video Input SetMainVideoSource ConnectorID: 1 SourceId: 4\x0D

Appendix B. Query Commands

Active Speaker PIP Position	xStatus Video ActiveSpeaker PIPPosition\x0D
Audio Volume	xStatus Audio Volume\x0D
Audio Volume Mute	xStatus Audio VolumeMute\x0D
Call ID	xStatus Call\x0D
Conference Do Not Disturb	xStatus Conference DoNotDisturb\x0D
DNS Server Address Status Server 1	xStatus Network 1 DNS Server 1 Address\x0D
DNS Server Address Status Server 3	xStatus Network 1 DNS Server 3 Address\x0D
H323 Gatekeeper Address Status	xStatus H323 Gatekeeper Address\x0D
H323 Gatekeeper Status	xStatus H323 Gatekeeper Status\x0D
IPv4 Address Status	xStatus Network 1 IPv4 Address\x0D
IPv4 Gateway Status	xStatus Network 1 IPv4 Gateway\x0D
IPv4 Subnet Mask Status	xStatus Network 1 IPv4 SubnetMask\x0D
Mute All Microphones	xStatus Audio Microphones Mute\x0D
Presentation Mode	xStatus Conference Presentation Mode\x0D
Self View	xStatus Video Selfview Mode\x0D
Self View Default Fullscreen Mode	xStatus Video Selfview FullscreenMode\x0D
Self View Position	xStatus Video Selfview PIPPosition\x0D
Standby	xStatus Standby State\x0D
Video Input Connector ID 1	xStatus Video Input Connector 1 SourceId\x0D